PRODUCT & INVENTORY MANAGEMENT SYSTEM

Prepared by Kai Carter

RAKIB HOSSEN

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PROJECT PLAN & FEATURE REQUIREMENTS

Project Type: Inventory and Small Business Management Software

Target Users: Grocery Stores, Confectionery Shops, and Stationery Shops

Initial Target Market: Local Retail Businesses in Khulna, Bangladesh

1. PROJECT OVERVIEW

The Product and Inventory Management System is designed to simplify daily business operations for small retail shop owners. Many shop owners rely on handwritten notebooks and manual calculations to manage inventory, track sales, calculate profits, and maintain customer credit records.

This system will provide a simple, centralized platform to manage products, monitor stock levels, record purchases and sales, track cash flow, and maintain accurate business records.

The primary goal is to reduce manual errors, prevent inventory losses, improve financial transparency, and help shop owners make better business decisions.

2. PROBLEM STATEMENT

Small retail businesses commonly face the following challenges:

2.1 Inventory Management Problems

- **Inaccurate Stock Records:** Shop owners often cannot determine the exact quantity of products available in their stores.
- **Expired Products:** Products may expire because purchase dates and expiry dates are not tracked properly.

- **Incorrect Selling Prices:** Shop owners may continue selling products at outdated prices, resulting in reduced profits or financial losses.
- **Supplier Payment Disputes:** Purchase payments, returned products, and outstanding supplier balances may not be recorded accurately.
- **Unclear Sales Trends:** Shop owners often lack reliable data to identify their best-selling products and purchasing patterns.

2.2 Financial Management Problems

- **Confusion Between Revenue and Profit:** Total sales revenue is often mistaken for actual profit, leading to poor financial decisions.
- **Manual Customer Credit Records:** Customer names, phone numbers, outstanding balances, and payment histories are maintained in notebooks.
- **Unrecorded Personal Expenses:** Money withdrawn from the shop for personal use is not always recorded separately.
- **Cash Reconciliation Issues:** Shop owners struggle to identify discrepancies between expected and actual cash balances.
- **Limited Sales Monitoring:** Without proper sales records and employee-level tracking, unauthorized discounts, unrecorded sales, and cash discrepancies can go unnoticed.

3. CORE FEATURES AND REQUIREMENTS

3.1 Product Management

The system will allow shop owners to manage product information efficiently.

- Add, edit, view, and delete product records.
- Store product names, categories, SKUs or barcodes, and units of measurement.
- Record purchase prices and selling prices.
- Track current stock quantities automatically.
- Set minimum stock thresholds and receive low-stock alerts.
- Update product prices when supplier prices change.
- Search and filter products by name, category, or barcode.
- View individual product details and stock movement history.

3.2 Inventory and Stock Management

The system will maintain accurate inventory records throughout daily business operations.

- Automatically increase stock when purchases are recorded.
- Automatically decrease stock when sales are completed.
- Record damaged, lost, or otherwise adjusted inventory.
- Track purchase dates and expiry dates for applicable products.
- Generate alerts for products approaching their expiry dates.
- Identify expired products and prevent their accidental sale where appropriate.
- Maintain stock adjustment and movement history.
- Support batch-level tracking for products with different expiry dates.

3.3 Sales Management

The system will provide a straightforward sales recording process.

- Record product sales with quantities and selling prices.
- Calculate subtotals, discounts, and total payable amounts automatically.
- Update inventory immediately after a successful sale.
- Support cash, mobile financial service, and other configured payment methods.
- Record partial payments and credit sales.
- Generate digital sales receipts or printable invoices.
- Maintain a complete sales history.
- Support sales returns and refunds with corresponding stock adjustments.
- Calculate sales totals for daily, weekly, and monthly periods.

3.4 Purchase and Supplier Management

The system will help shop owners track incoming stock and supplier transactions.

- Record product purchases, quantities, and purchase prices.
- Maintain supplier names, contact details, and transaction histories.
- Track payments made to suppliers.
- Record outstanding supplier balances.
- Track returned products and purchase returns.
- Maintain purchase invoices and transaction dates.

- Calculate the remaining amount payable to each supplier.
- View supplier-wise purchase reports.

3.5 Customer and Credit Management

The system will maintain organized customer records and outstanding credit balances.

- Add customer names and phone numbers.
- Record credit sales against specific customers.
- Track outstanding balances for each customer.
- Record partial and full repayments.
- Maintain customer payment histories.
- Display total outstanding customer credit.
- Generate reminders for overdue payments.
- Provide quick access to customer transaction details.

Future enhancement: Automated SMS or messaging reminders, subject to the availability of suitable services and customer consent.

3.6 Cash and Financial Management

The system will help shop owners understand where their money comes from and where it goes.

- Record opening cash balances.
- Track cash received from sales and customer repayments.
- Record supplier payments and other business expenses.
- Record owner withdrawals and personal expenses separately.
- Calculate expected closing cash balances.
- Compare expected cash with the actual cash counted at closing.
- Highlight cash discrepancies.
- Track daily income and expenses.
- Generate financial summaries for selected periods.

3.7 Profit and Loss Tracking

The system will provide a clearer view of business profitability.

- Calculate sales revenue automatically.

- Calculate the cost of goods sold (COGS).
- Calculate gross profit from completed sales.
- Record operating expenses separately.
- Calculate estimated net profit after applicable expenses.
- Display profit and loss summaries by day, week, and month.
- Identify products generating the highest sales and gross profit.
- Track changes in purchase costs and selling prices.

Important: Sales revenue, cash balance, gross profit, and net profit will be treated as separate financial measures.

3.8 Dashboard and Reports

The dashboard will provide an overview of the shop's current business condition.

- Total sales for the current day.
- Estimated gross profit and net profit.
- Current inventory value.
- Low-stock product alerts.
- Products approaching their expiry dates.
- Total outstanding customer credit.
- Total outstanding supplier payments.
- Daily cash summary.
- Best-selling products.
- Sales trends and expense summaries.

Reports should be filterable by date range and exportable to common formats, such as PDF or CSV, where appropriate.

4. USER INTERFACE AND ACCESSIBILITY

The system should be easy to use for shop owners with limited technical experience.

- Mobile-friendly and responsive interface.
- Simple navigation and clearly labeled actions.
- Fast product search and sales entry.
- Readable dashboard summaries.
- Bangla-language interface support.

- Minimal steps for common tasks.
- Clear validation messages and error notifications.
- Secure login and role-based access where multiple employees use the system.

5. SECURITY AND DATA RELIABILITY

The system must protect business records and reduce the risk of accidental data loss.

- Secure user authentication.
- Role-based permissions for owners and employees.
- Validation of product, sales, and payment data.
- Transaction-safe inventory updates.
- Protection against duplicate or incomplete transactions.
- Regular database backups and recovery procedures.
- Audit logs for important actions and stock adjustments.
- Secure handling of customer contact information.

6. FUTURE ENHANCEMENTS

The following features may be introduced after the core system is stable.

- Barcode scanning for faster sales entry.
- Receipt printing and thermal printer support.
- Multiple shop or branch management.
- Advanced employee performance reports.
- Automated customer payment reminders.
- Offline sales recording with synchronization.
- Supplier order recommendations based on sales trends.
- Advanced sales forecasting and inventory analytics.
- Integration with accounting software and payment services.

7. MINIMUM VIABLE PRODUCT (MVP)

The first release should focus on the features that solve the most urgent problems without making the system unnecessarily complicated.

Phase 1 — Essential Features

1. Product and category management.
2. Stock tracking and automatic inventory updates.
3. Purchase and sales recording.
4. Customer credit management.
5. Supplier payment tracking.
6. Daily cash reconciliation.
7. Basic profit and loss calculations.
8. Low-stock and expiry alerts.
9. Dashboard with essential reports.
10. Secure authentication and database backup.

Phase 2 — Business Improvements

1. Barcode support and printable receipts.
2. Advanced reports and data exports.
3. Employee activity and audit logs.
4. Automated payment reminders.
5. Enhanced sales and inventory analytics.

Phase 3 — Advanced Capabilities

1. Multi-branch management.
2. Offline operation and synchronization.
3. Sales forecasting and intelligent stock recommendations.
4. Third-party accounting and payment integrations.

8. EXPECTED OUTCOMES

The proposed system is intended to help small shop owners:

- Maintain accurate product and inventory records.
- Reduce losses from expired, damaged, or missing products.
- Avoid selling products at outdated prices.
- Keep customer credit and supplier payment records organized.
- Separate business revenue, expenses, cash, and profit.
- Identify cash discrepancies more quickly.
- Make better purchasing and pricing decisions.
- Save time spent on manual calculations and recordkeeping.

9. PROJECT SUCCESS CRITERIA

The system's effectiveness should be evaluated using measurable outcomes.

- Accuracy of inventory records compared with physical stock.
- Accuracy of expected versus actual cash balances.
- Reduction in time required for daily bookkeeping.
- Accuracy of customer credit and supplier balances.
- Timeliness of low-stock and expiry notifications.
- Reliability of sales, return, and purchase records.
- Ease of use for shop owners and employees.

10. CONCLUSION

The Product and Inventory Management System aims to provide a practical, accessible, and reliable business management solution for small retail shops.

Rather than simply digitizing a traditional notebook, the system will connect inventory, purchases, sales, customer credit, supplier payments, and financial records into one consistent workflow.

The initial focus will be on simplicity, accuracy, and mobile accessibility. Additional automation and advanced reporting can be introduced as the system evolves and real shop owners provide feedback.

Project Vision: To make everyday business management easier, more transparent, and more reliable for small retail shop owners.

app usage.